SautiLedger — ASR Benchmark for Code-Switched Financial Speech

Deep Learning Indaba 2026 · MLC (Africa) × Intron Agentic Voice AI Challenge · David Agbolade

Models: Intron Sahara-v2 (cloud API) · Whisper large-v3 (local) · Whisper small (local, frontier-slot substitute) · Corpus frozen before first run, sha256 `d68d9044…`

Summary — final (v3) scoring, all models, both tiers

Tier A: SautiLedger market-speech corpus (15 clips, Nigerian Pidgin/Yoruba/English, parse ground truth)

Model	WER (norm)	WER (raw)	Numeric acc	Txn exact	Amount safe	Amount corrupted
sahara-v2	**60.2%**	**70.4%**	**73%**	**47%**	**93%**	**7%**
whisper-large-v3	96.3%	105.4%	67%	27%	100%	0%
whisper-small	85.5%	95.0%	73%	13%	100%	0%

Tier B: AfriSwitch wild code-switched speech (40 clips; WER only — no parse ground truth)

Model	WER (norm)	WER (raw)
sahara-v2	**42.4%**	**56.2%**
whisper-large-v3	76.4%	85.1%
whisper-small	77.6%	87.3%

> **Headline:** Sahara-v2 is the only model that produces a usable financial ledger from this speech (47% exact transactions vs 13–27%), and leads wild code-switched WER by ~1.8×. Its one weakness is instructive: numeric normalisation can flatten native money grammar into plausible-but-wrong figures — the exact failure class the agent's clarify layer exists to catch (see p.2). Whisper's 0% corruption is not safety: its output is largely unparseable, so nothing reaches the ledger at all.

Per-model assessment

Intron Sahara-v2 — *Pros:* built for this speech; best on market entities (derica, garri), Yoruba numerals, and numeric-heavy utterances; only model with production-viable Pidgin, Swahili and Hausa; strongest downstream transaction accuracy. *Cons:* cloud-only as tested (offline deployment exists but was not under test); digit normalisation collapses reduplicated money ("two two fifty" → "250"), producing plausible transcripts that need a downstream guard; one deletion-class error survives ("ten thousand" → "thousand").

Whisper large-v3 — *Pros:* strong general model; runs fully offline; reasonable on clear accented English. *Cons:* anglicises Pidgin grammar (perfective "don sell" → negated "don't sell" — a meaning inversion observed in the wild); silently *translates* Swahili input into English instead of transcribing it; mangles market vocabulary; effectively unusable downstream (27% exact).

Whisper small — *Pros:* lightweight, offline, fast; honest floor for a local baseline. *Cons:* output on code-switched market speech is frequently noise ("Set 1, 4, 5, 6..."); its low corruption rate reflects unparseability, not fidelity. Used only because no frontier API key was available; a weaker baseline than GPT-4o/Gemini, stated as such.

Data

Tier	Source	Languages	Clips	Audio	Ground truth
A	SautiLedger corpus — one adult male Nigerian speaker (the developer), phone-recorded market utterances	Nigerian Pidgin / Yoruba / English mix	15	~1.1 min	Verbatim transcript + expected transaction parse per clip
B	AfriSwitch (intronhealth/AfriSwitch, HuggingFace; gated, CC BY-NC-SA 4.0)	Pidgin-En (16), Swahili-En (12), Yoruba-En (6), Hausa-En (6)	40	8.2 min	Dataset transcription field (no parse labels)

Tier-A utterances were drafted with AI assistance and **corrected by a native Nigerian Pidgin/Yoruba speaker** before recording (e.g. "five k five" was AI-Pidgin; the real form is "five thousand five"). Swahili/Hausa companion cases remain non-native drafts pending validation and were not recorded. The corpus manifest was frozen (sha256) before the first transcription; no clip was added or dropped after seeing results.

Preprocessing — including an audio-conversion correction

- Tier-B audio is used exactly as published (original bytes, no re-encode). Tier-A phone recordings (48 kHz AAC) are converted to 16 kHz mono PCM WAV.
- **Correction:** the initial conversion had a resampler bug that time-stretched every Tier-A clip by ~1.26× (frame padding copied as samples). It was detected mid-evaluation via an A/B against the vendor's own web UI, root-caused, fixed, and **Tier A was re-measured for all three models (v3)** from re-converted audio. All models improved; pre-correction scorings are preserved. Tier B was unaffected. Details on p.3.
- Sahara calls: documented sync endpoint, language codes pcm / sw / yo / ha (API-validated). Whisper: faster-whisper, int8, language=en. Cloud requests are routed through the app's egress logger — the benchmark obeys the same transmission-audit rule as the product.

Metrics — and why transaction accuracy matters for financial speech

- **WER** (word error rate), reported normalised (lowercase, punctuation stripped; jiwer-standard) and raw. The upstream Intron benchmarking repo reports dual WER but does not publish its normaliser, so ours is stated rather than claimed identical.
- **Numeric accuracy** — did every monetary amount and quantity in the ground truth survive transcription in a recoverable form ("5.5k" ≡ "five thousand five" ≡ 5500)?
- **Transaction accuracy** — each model's raw transcript is fed through SautiLedger's deterministic, native-speaker-validated grammar, and the resulting parse is scored three ways: **exact** (full parse correct), **amount-safe** (amount correct *or* the system asked a clarifying question — asking is safe), **amount-corrupted** (a wrong amount would have been written to the ledger).

> **Why this matters:** WER treats every word equally; money does not. "Customer take 2 pint of Dairy 250" scores decently on WER — five content words survive — yet it silently halves a customer's bill (truth: 250 *each*, 500 total). Conversely, Sahara's Tier-A WER is *inflated* by digit renderings that are semantically correct ("5,500" counted as errors against spoken "five thousand five"). For financial speech, transcription accuracy is necessary but not sufficient; the amount-corrupted column — the number that must be ~0 — measures what actually reaches someone's money records. It is also where engineering can intervene: a distributive-guard added to the parser after measurement halved Sahara's corruption (13% → 7%) by converting flattened amounts into clarifying questions.

Per-language results (WER, normalised — v3)

Language (pair)	n	sahara-v2	whisper-large-v3	whisper-small
Pidgin — market commands (Tier A)	15	60.2%	96.3%	85.5%
Pidgin-English — wild (Tier B)	16	33.3%	33.5%	33.7%
Swahili-English (Tier B)	12	33.6%	118.9%	116.2%
Yoruba-English (Tier B)	6	91.6%	90.9%	94.4%
Hausa-English (Tier B)	6	35.3%	91.4%	100.6%

Sahara leads by ~3.5× on Swahili (whisper's WER >100% because it *translates* to English) and ~2.7× on Hausa. Wild conversational Pidgin-English is near-parity — it is lexically close to English; the differentiation appears exactly where financial content lives: dense numerals, market units, and short command speech (Tier A). Yoruba-heavy speech is hard for every model tested. Small n throughout — directional, not population-level.

Qualitative findings

```
truth: I don sell three derica of rice five thousand five
sahara: "I don sell 3 derica of rice 5,500." · whisper-large: "I don't sell three Delica or Bryce 5005."
(meaning inversion + corrupted amount)
```

```
truth: customer take two paint rubber of garri two two fifty (= ₦250 each — native reduplication)
sahara: "Customer take 2 print of garri 250." — the distributive collapsed to one figure; caught downstream
by the clarify guard
```

```
truth (Swahili): kile kiwango kinachotakiwa cha kulipa dividends
sahara: near-verbatim · whisper-large: "If I don't have money, I won't be able to pay the dividend."
(translation, not transcription)
```

- **The corruption inversion:** the weakest model posts the lowest corruption — unparseable noise never reaches the ledger — while the strongest model's errors are plausible enough to survive parsing. Downstream safety must be engineered, not assumed from accuracy.
- **Meaning inversion in the wild:** both whisper models turned perfective "I don sell" into negated "I don't sell", flagged automatically by the harness.
- **Native grammar is load-bearing:** the corpus itself required native-speaker correction ("two two fifty" is unambiguously distributive, not ambiguous) — the same gap this product exists to close.

Methodology note — configuration & correction history

Three scorings are preserved. **v1→v2** (same audio): two documented post-freeze grammar amendments — a digit-twin rule ("5k 5" = spoken "five thousand five") and a flattened-distributive guard (quantity ≥ 2 + single bare numeral → clarify) — moved Sahara's corruption 20%→13% with every movement reported. **v2→v3** (corrected audio): after an A/B against the vendor web UI exposed better transcripts for identical clips, the fault was isolated to the benchmark's own audio conversion (1.26× time-stretch); the language configuration was verified correct and API-validated throughout. Tier A was re-converted and re-measured for all models: Sahara WER 71.6→60.2, numeric 60→73, corrupted 13→7; whisper-large corrupted 7→0; whisper-small WER 119.4→85.5. Raw transcripts are cached; all numbers are reproducible from the frozen corpus without new API spend.

Citations & licences

AfriSwitch — Intron Health, huggingface.co/datasets/intronhealth/AfriSwitch (CC BY-NC-SA 4.0; used for evaluation only, fetched at run time, never redistributed, not used to build the product). · Olatunji et al., "AfriSpeech-200: Pan-African Accented Speech Dataset for Clinical and General Domain ASR", TACL 2023. · Intron Sahara-v2 API, docs.voice.intron.io. · github.com/intron-innovation/Intron-Multimodal-Benchmarking (WER reporting conventions). · Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision" (Whisper), 2022. · Full report with per-clip transcripts: bench/results/REPORT.md in the project repository.